

Multimodal Sequencing for POS Tagging

Baraa Al Jorf
baj321@nyu.edu

Sophie Juco
smj490@nyu.edu

Joel Marvin Tellis
jt4680@nyu.edu

Nirman Taterh
nt2613@nyu.edu

Abstract

Sequencing unordered events is a critical facet of comprehension and reasoning in real-world tasks, playing a pivotal role in applications such as task planning and multi-source instruction summarization. This paper explores the integration of image data to enhance Part-of-Speech (POS) tagging within the domain of Natural Language Processing (NLP). Drawing inspiration from the study of understanding multimodal procedural knowledge through the sequencing of instructional manuals, our research investigates the extent to which image data can contribute to POS tagging tasks. The results expose a disparity in efficacy between state-of-the-art models and human performance, inspiring the introduction of pretraining techniques attuned to sequential contexts. These methods leverage the sequential alignment attributes inherent in both text and images, leading to significant enhancements in the perfect match ratio. This approach effectively tackles the complexities associated with multimodal sequencing tasks.

1 Introduction

Instructions serve as pivotal resources for agents to learn and execute complex tasks, necessitating the comprehension of multiple steps often presented in a non-sequential order. While existing work has addressed the sequencing of unordered texts, real-world tasks are further complicated by the inclusion of multimodal information, combining textual descriptions with images. Our work extends this exploration by focusing on the integration of image data to improve Part-of-Speech (POS) tagging within the realm of Natural Language Processing (NLP).

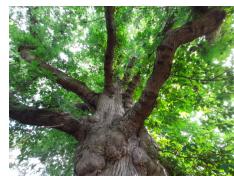
Motivated by a study on understanding multimodal procedural knowledge, we aim to investigate the potential of leveraging image data for effective POS tagging task. Consider the following sentences:

1. The dogs **bark**. (POS tag for bark : VERB)
2. The tree has brown **bark**. (POS tag for bark : NOUN)
3. Add a **minute** to the timer. (POS tag for minute : NOUN)
4. A krill is like a **minute** shrimp. (POS tag for minute : ADJECTIVE)

The example above demonstrates the idea that the same grapheme can be assigned different POS tags based on the context in which it is used in the sentences. This is often the case with homographs which are words with the same spelling but different meanings. Although these are very simple examples, they indicate that handling such cases in POS tagging requires an efficient model structure trained with extensive data. Now consider the following inputs where the sentence is also associated with a related image.



The dogs **bark**.



The tree has brown **bark**.



Add a **minute** to the timer.



Krill are like **minute** shrimps.

The associated images provided more information about the context in which a particular word is being used in the given sentence. For our task, we utilize the RecipeQA dataset, scraped of the instructables.com website, to annotate paired image to word to tag training and testing corpus. We aim to leverage this additional visual information from the associated image to improve the POS tagging for sentences.

Our preliminary studies indicate that models struggle to effectively utilize multimodal information. To bridge this gap, we propose pretraining techniques that exploit the sequential alignment properties of both textual and visual modalities. We utilise powerful architectures such as VGG16 for pre-processing the image data, and implore a mutlimodal neural network architecture to demonstrate the effectiveness of our model in assigned tagging task.

Our contributions can be summarized as follows:

- Creation of POSnet, a multimodal architecture that utilizes joint fusion on text embeddings and VGG16 extracted features to assign POS tags
- Annotation of Word to tag to image dataset using Online available RecipeQA dataset.

2 Related Work

Information quantification, a concept pivotal in understanding the varying degrees of data conveyed through different mediums, plays a crucial role in fields like Natural Language Processing (NLP) and computer vision. This concept stems from the understanding that different modalities—text and images, for instance—carry inherently diverse amounts of information. A key theoretical foundation in this area is Allan Paivio’s Dual-Coding Theory. This theory suggests that visual and verbal information are processed in separate channels in the human mind, leading to distinct representations. This concept is instrumental in understanding how images and text contribute differently to cognitive processes and learning. Furthermore, it provides a basis for considering how the integration of these modalities can enhance POS tagging,

particularly in handling homographs or context-dependent words(Paivio, 2021).

Significant research has been conducted to quantify the information content in images versus text. For example, () work on the Cognitive Theory of Multimedia Learning suggests that a combination of visual and verbal explanations promotes better understanding and retention. This theory supports the idea that the amalgamation of text and images in POS tagging could significantly improve accuracy and comprehension. Studies focusing on multimodal learning and comprehension also play a crucial role in this discussion. They highlight how the integration of text and images can enhance learning experiences, a concept aligned with Richard E. Mayer’s findings. These studies provide empirical backing for our proposed multimodal approach, which leverages the comprehensive context offered by images alongside the specificity of textual data.

These connections are examined in work like Wu et al. (2021) where language and multimodal models are explored utilizing widely-used online instructional manuals. The work demonstrates the beneficial impact of both multimodality and novel sequence-aware pretraining approach on multimodal sequencing and offers valuable insights into multi-reference annotations for the sequencing task, aiming to stimulate future research in this domain. Pham et al. (2018) proposes two unsupervised learning methods for joint multimodal representations in sentiment analysis. The paper explores Seq2Seq Modality Translation Model and Hierarchical Seq2Seq Modality Translation Model demonstrating improved performance on multimodal sentiment analysis, particularly in the Bimodal case. The research addresses the gap in unsupervised representation learning, showcasing the effectiveness of learned multimodal representations for downstream tasks. Shen et al. (2021) investigates the benefits of using CLIP (Contrastive Language-Image Pre-training) as a visual encoder in Vision-and-Language models. Additionally, Bugliarello et al. (2021) explores large-scale pretraining and task-specific finetuning methods, specifically focusing on vision and language BERTs. The authors categorize these models into single-stream and dual-stream encoders and present a unified theoretical framework. The paper aims to provide a better understanding of massive models in the intersection of vision and language. Yagcioglu et al. (2018) introduces RecipeQA, a multi-

modal dataset designed for the comprehension of cooking recipes. With approximately 20,000 instructional recipes, each containing titles, descriptions, and aligned images, the dataset includes over 36,000 automatically generated question-answer pairs. The authors propose a set of comprehension and reasoning tasks that require joint understanding of both images and text, emphasizing the temporal flow of events and the interpretation of procedural knowledge. Preliminary results suggest that RecipeQA serves as a challenging benchmark for evaluating machine comprehension systems, particularly in the context of cooking recipes. Furthermore, [Nguyen and Daumé III \(2019\)](#) introduce an interactive photo-realistic simulator where a mobile agent fulfills object-finding tasks by requesting and interpreting natural language and vision assistance from simulated human assistants called ANNA (Automatic Natural Navigation Assistants). An end-to-end model designed by [Tsai et al. \(2019\)](#) addresses challenges in modeling multimodal human language time-series data, specifically dealing with inherent data non-alignment and long-range dependencies between modalities. The proposed model employs directional pairwise crossmodal attention to capture interactions between multimodal sequences without explicit data alignment. [Yang et al. \(2021\)](#) introduces the Visual Goal-Step Inference, which involves choosing an image that represents a plausible step toward a given textual goal. The authors create a new dataset for VGSI using wikiHow articles, consisting of over 770,000 images representing human actions. State-of-the-art multimodal models struggle with VGSI, indicating its challenging nature.

3 Problem Definition

As such, it is imperative to see if the joint representation of information from textual and visual modalities can help in adding context to text - in a manner that facilitates the grammatical understanding of the text. The task at hand is therefore, to perform POS tagging on a pair of inputs consisting of two types of contents: a textual description T_i of tokens and correlated image(s) I_i .

4 Methodology

To tackle this task, we propose a mutlimodal architecture, that utilizes an image feature extractor, and a joint fusion-module.

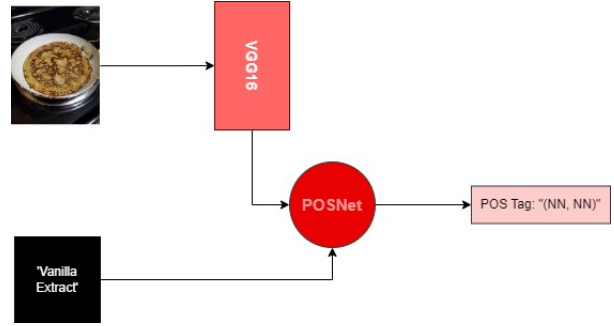


Figure 1: Overview of the overall architecture. POSNet is a two-layer neural network.

4.1 Prediction Task

The goal is to predict the POS tag corresponding to each word-image pair in the dataset. This is done through a POSNet neural network that is trained with a cross-entropy loss.

4.2 Dataset

To demonstrate the effectiveness of multimodal tagging, we use a dataset where images are paired to text. The RecipeQA dataset presented in the work discussed above by [Reif et al., 2022](#) can be easily manipulated into an annotated dataset of the desired form. It comprises a list of recipes divided by steps, where steps are paired with images. We use this dataset to annotate our own text to image dataset. We divide each step into words, and pair each one of those words to the image originally paired to the step. The images in our dataset are processed using a pretrained VGG16 model with imagenet weights. The extracted features are concatenated to our text representations in a joint-fusion architecture. We then use an "averaged perceptron tagger" to determine the truth labels for the dataset, which we then compare against our predicted labels using our mutlimodal tagger.

4.3 Multimodal Predictions:

The core of the method is the POSNet neural network. It is a feedforward neural network consisting of two fully connected (linear) layers. It takes as input both the text features (one-hot encoded word vectors) and image features concatenated together using the fusion module. The first layer has 256 hidden units with a ReLU activation function, and the second layer produces softmax predictions for POS tags. The model architecture is shown in figure 1 below.

5 Baseline Results

To determine the effectiveness of our model, we compare its metrics to those of state-of-the-art tagging algorithms. With the predicted POS tags, we generate the precision, recall, and F1 scores. We compare those metrics against text-only POS tagging systems, namely unigram and bigram taggers. The final evaluation we conduct is a manual evaluation of a subsection of the test data to compare the model’s results to human generated results. The results are all shown in table 1.

5.1 Unigram Tagging

Unigram POS tagging relies on the analysis of individual words in isolation, assigning each word a part-of-speech tag based solely on its own characteristics and context within the sentence. This method is relatively straightforward and computationally efficient but may struggle with disambiguating words that have multiple possible POS tags depending on the context.

5.2 Bigram Tagging

Bigram POS tagging considers pairs of consecutive words in a sentence, taking into account the relationship between a word and its immediate neighbor. This approach can capture contextual information more effectively and address some of the ambiguity challenges faced by unigram tagging. NLTK’s BigramTagger model is trained on the same training split of the data and predictions are generated using the same test split of the data.

5.3 Manual Human Evaluation

[hbt!] Due to the size of the data, we conduct the manual human evaluation over a smaller subsection of the data. In particular, we perform the manual evaluation on the first five recipes in the test set which, combined, contains 1693 tokens. The selected data are given a POS tag by hand and the results are compared against the expected data to generate precision, recall, and F1 evaluation metrics. The POS tagging rules followed are the Penn Treebank POS tags which are defined by [Marcus et al. \(1993\)](#) and the accompanying Figure 2 below.

6 Discussion

The multimodal model demonstrates the highest precision (0.9681) and a significantly improved recall (0.8368) compared to the NLTK Unigram and Bigram models. This suggests that the multimodal

The Penn Treebank POS tagset.

1. CC	Coordinating conjunction	25. TO	to
2. CD	Cardinal number	26. UH	Interjection
3. DT	Determiner	27. VB	Verb, base form
4. EX	Existential <i>there</i>	28. VBD	Verb, past tense
5. FW	Foreign word	29. VBG	Verb, gerund/present participle
6. IN	Preposition/subordinating conjunction	30. VBN	Verb, past participle
7. JJ	Adjective	31. VBP	Verb, non-3rd ps. sing. present
8. JJR	Adjective, comparative	32. VBZ	Verb, 3rd ps. sing. present
9. JJS	Adjective, superlative	33. WDT	<i>wh</i> -determiner
10. LS	List item marker	34. WP	<i>wh</i> -pronoun
11. MD	Modal	35. WP\$	Possessive <i>wh</i> -pronoun
12. NN	Noun, singular or mass	36. WRB	<i>wh</i> -adverb
13. NNS	Noun, plural	37. #	Pound sign
14. NNP	Proper noun, singular	38. \$	Dollar sign
15. NNPS	Proper noun, plural	39. .	Sentence-final punctuation
16. PDT	Predeterminer	40. ,	Comma
17. POS	Possessive ending	41. :	Colon, semi-colon
18. PRP	Personal pronoun	42. (	Left bracket character
19. PPS	Possessive pronoun	43.)	Right bracket character
20. RB	Adverb	44. "	Straight double quote
21. RBR	Adverb, comparative	45. '	Left open single quote
22. RBS	Adverb, superlative	46. "	Left open double quote
23. RP	Particle	47. '	Right close single quote
24. SYM	Symbol (mathematical or scientific)	48. "	Right close double quote

Figure 2: Penn Treebank POS tagging rules.

Tagging System	Precision	Recall	F1
Multi-modal	0.9681	0.8368	0.8937
NLTK Unigram	0.9587	0.8213	0.8847
NLTK Bigram	0.9619	0.3459	0.5088
Manual (1693 tokens)	0.7974	0.7832	0.7902

Table 1: Evaluation metrics of the model and the comparison evaluations.

approach is more effective in correctly identifying relevant POS tags (high precision) while also capturing a larger proportion of the necessary information (higher recall). The enhanced performance of the multimodal model can be attributed to its ability to process and integrate diverse types of information. By combining textual and visual data, the model gains access to a richer, more contextual data set. This aligns with the principles of information quantification, where the amalgamation of different data types provides a more comprehensive understanding of the context, leading to more accurate POS tagging.

The NLTK Unigram model, while showing high precision, falls short in recall, indicating its limitations in capturing the full scope of necessary information for POS tagging. This is typical of unigram models that rely solely on individual word analysis, often missing contextual nuances. The NLTK Bigram model, despite having a slightly higher precision than the Unigram model, significantly lags in recall. This suggests that while it is slightly better at avoiding false positives, it fails to capture a substantial amount of relevant information, likely due to its dependency on adjacent word pairs which may not always provide sufficient context. The manual evaluation, conducted on a subset of the data, shows considerably lower precision and recall scores. This highlights the challenges humans face in consistently and accurately tagging

POS, especially in complex or ambiguous contexts. It underscores the advantage of automated systems, particularly multimodal models, in handling large and diverse datasets with higher consistency and accuracy.

Given more time and resources for this project, we would have liked to have expanded our dataset to incorporate the WikiHow data used in the paper by Wu et al. (2021). This additional data would have not only given us more train, validation, and test data to work with and train our models on but it would have also provided us with a larger diversity of text topics. The RecipeQA data used consist of various food preparation instructions and images. The WikiHow data consist of instructions and related images that cover a much broader field. The intent with incorporating the WikiHow data would be to make the model better generalized and the results reproducible for various datasets. Additionally, given more time, we would have generated more precise true POS tags for the tokens in the dataset. From manually tagging a portion of the test data, it is clear that the grammar of the instructions is not always correct and that the chunking and tokenization methods did not properly separate all tokens. For example, a single token would include two words together (ex. "carrotAdd") or a word and a punctuation mark (ex. "oven."). The text data of RecipeQA also contained many symbols like hyphens or asterisks that were labeled with word POS tags instead of symbol POS tags. These discrepancies could also be the reason the resulting manual tags produced a lower precision, recall, and F1 scores than the unigram tagger.

For future iterations of this project, expanding on the evaluation comparison models could be useful in understanding the use of multimodal image and text sequencing and POS tagging. In our current set up, we compare our multimodal tagger to text-only unigram and bigram tagging models. In future versions, the model could be compared to various, more complex n-gram models with the thought that they would carry more information much like our multimodal model does by incorporating image data. Additionally, if another study exists that incorporates multimodal POS tagging using another mode of data other than images - perhaps audio data - it would be interesting to compare the results of one additional mode of data (images) against another (audio) and evaluate which can be better used in a machine learning model to predict POS

tags.

7 Conclusion

the multimodal model's superior performance can be attributed to its ability to effectively quantify and integrate information from multiple sources. This aligns with the principles of information quantification, where leveraging diverse data types leads to a more holistic understanding and processing of information. The results thus advocate for the multimodal approach in POS tagging, emphasizing its ability to transcend the limitations of text-only models and offering a more robust and accurate solution.

8 Reproducibility

We share our repository with data processing and code on GitHub. All distributed experiments were conducted on NYU Greene HPC, with AMD EPYC 7742 64-Core Processor with 4GB of RAM per allocated core (128GB per node) with 10 nodes in total. All code should be accesible on this link:

<https://github.com/BaraaAlJorf/POSnet>

References

- Emanuele Bugliarello, Ryan Cotterell, Naoaki Okazaki, and Desmond Elliott. 2021. [Multimodal pretraining unmasked: A meta-analysis and a unified framework of vision-and-language BERTs](#). *Transactions of the Association for Computational Linguistics*, 9:978–994.
- Mitchell P. Marcus, Beatrice Santorini, and Mary Ann Marcinkiewicz. 1993. [Building a large annotated corpus of English: The Penn Treebank](#). *Computational Linguistics*, 19(2):313–330.
- Khanh Nguyen and Hal Daumé III. 2019. [Help, anna! visual navigation with natural multimodal assistance via retrospective curiosity-encouraging imitation learning](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 684–695, Hong Kong, China. Association for Computational Linguistics.
- Allan Paivio. 2021. [Dual-coding theory](#). *CoRR*, abs/2110.08486.
- Hai Pham, Thomas Manzini, Paul Pu Liang, and Barnabás Póczós. 2018. [Seq2Seq2Sentiment: Multimodal sequence to sequence models for sentiment analysis](#). In *Proceedings of Grand Challenge and Workshop on Human Multimodal Language (Challenge-HML)*,

pages 53–63, Melbourne, Australia. Association for Computational Linguistics.

Emily Reif, Daphne Ippolito, Ann Yuan, Andy Coenen, Chris Callison-Burch, and Jason Wei. 2022. [A recipe for arbitrary text style transfer with large language models](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 837–848, Dublin, Ireland. Association for Computational Linguistics.

Sheng Shen, Liunian Harold Li, Hao Tan, Mohit Bansal, Anna Rohrbach, Kai-Wei Chang, Zhewei Yao, and Kurt Keutzer. 2021. [How much can CLIP benefit vision-and-language tasks?](#) *CoRR*, abs/2107.06383.

Yao-Hung Hubert Tsai, Shaojie Bai, Paul Pu Liang, J. Zico Kolter, Louis-Philippe Morency, and Ruslan Salakhutdinov. 2019. [Multimodal transformer for unaligned multimodal language sequences](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 6558–6569, Florence, Italy. Association for Computational Linguistics.

Te-Lin Wu, Alexander Spangher, Pegah Alipoormolabashi, Marjorie Freedman, Ralph M. Weischedel, and Nanyun Peng. 2021. [Understanding procedural knowledge by sequencing multimodal instructional manuals](#). *CoRR*, abs/2110.08486.

Semih Yagcioglu, Aykut Erdem, Erkut Erdem, and Nazli Ikizler-Cinbis. 2018. [RecipeQA: A challenge dataset for multimodal comprehension of cooking recipes](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 1358–1368, Brussels, Belgium. Association for Computational Linguistics.

Yue Yang, Artemis Panagopoulou, Qing Lyu, Li Zhang, Mark Yatskar, and Chris Callison-Burch. 2021. [Visual goal-step inference using wikiHow](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 2167–2179, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.